Evaluating Piecewise and Absolute-Value Functions

Absolute-value values, choosing the right piecewise rule, and working backwards to the input

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- $|a|$ is the **distance** of a from zero, so it is never negative: $|-5| = 5$. Work out what is inside the bars first, then take the distance, then do everything outside.
- A **piecewise** function is several rules with a condition on each. Before you compute anything, check which condition your input satisfies — that is the whole skill.
- Watch the difference between $<$ and $\leq$. Exactly one rule may claim a boundary value.
- Show the substitution, not just the answer: $f(7) = |7 - 3| + 2 = 4 + 2 = 6$.

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1. Complete the table for $f(x) = |x - 3| + 2$.

x	$f(x)$
0	________
3	________
7	________

2. For $g(x) = -2|x + 1| + 5$, find $g(-4)$.

Answer: ________________

3. A piecewise function is defined by

$$f(x) = \begin{cases} 3x+1, & x < 2 \\ x^2 - 3, & x \geq 2 \end{cases}$$

(a) Find $f(0) =$ ________ $f(2) =$ ________ $f(5) =$ ________

(b) In the space below, explain why $x = 2$ uses the second rule and not the first.

4. For $f(x) = |x-3| + 2$, find *every* value of x for which $f(x) = 7$.

Answer: ________________

5. A room thermostat is set to 68 degrees Fahrenheit. How far the actual temperature t is from the setting is measured by $E(t) = |t - 68|$ degrees.

(a) The room reaches 74 degrees. Find $E(74) =$ ________

(b) Later it falls to 63 degrees. Find $E(63) =$ ________

6. A car park charges by the hour. For a stay of h hours the cost in dollars is

$$C(h) = \begin{cases} 4 + 2h, & 0 \leq h \leq 3 \\ 10 + 5(h - 3), & h > 3 \end{cases}$$

Find the cost of a stay of 2 hours, of 3 hours, and of 6 hours.
$C(2) =$ ________ $C(3) =$ ________ $C(6) =$ ________

7. For $g(x) = -2|x + 1| + 5$, find *every* value of x for which $g(x) = -1$.

Answer: ________________

8. A piecewise function is defined by

$$p(x) = \begin{cases} x+4, & x \le -1 \\ 2-x^2, & x > -1 \end{cases}$$

$p(-1) =$ ________ $p(-3) =$ ________ $p(2) =$ ________

9. Use the function f from Problem 3 again:

$$f(x) = \begin{cases} 3x+1, & x < 2 \\ x^2-3, & x \ge 2 \end{cases}$$

(a) Solve $3x + 1 = 13$. (b) Solve $x^2 - 3 = 13$.
(c) Which of the values you found are genuine solutions of $f(x) = 13$? Explain your decision using the condition attached to each rule.

Answer: ________________

10. Synthesis challenge. A straight-line model $y = x + 1$ and a quadratic model $y = x^2 - 1$ are compared.

(a) Solve the system to find every point where the two models give the same output. Write the points as ordered pairs.

(b) Explain what those points mean about the two models, and why checking one input would not have been enough.

Answer: ____________________